MainFaye_Isabel_2026_endassignment.pdeSetup, game loop, user input, character selection
TankBackgroundbackground.pdeWater gradient + Perlin noise sand waves
Coralcoral.pdeMass-spring-damper rotational system
Fishfish.pdeFlocking (cohesion, alignment, separation + flee)
Jellyfishjellyfish.pdePerlin noise movement, sine wave pulsing, spawns particles
Tentacletentacles.pdeChain of nodes with Perlin noise water sway
ParticleSystemjellyfish_particles.pdeManages ArrayList of Particle objects
Particlejellyfish_particles.pdeSingle glowing particle with lifetime
Start Screenchoose_animal_background.pdeCharacter selection UI (functions, not a class)

Main ──► creates ──► TankBackground
Main ──► creates ──► ArrayList<Coral>
Main ──► creates ──► ArrayList<Fish>
Main ──► creates ──► Jellyfish
Main ──► creates ──► ParticleSystem

Jellyfish ──► uses ──► ParticleSystem (global reference ps)
Jellyfish ──► contains ──► Tentacle[7] (composition)

ParticleSystem ──► contains ──► ArrayList<Particle> (composition)

Fish ◄── Main passes (charX, charY) for flee behavior
Coral ◄── Main passes waterForce (Perlin noise)

MainSketch

-gameState: int
-selectedCharacter: PImage
-charX: float
-charY: float
-charSpeed: float
-upPressed: boolean
-downPressed: boolean
-leftPressed: boolean
-rightPressed: boolean
-generator: Random
-bg: TankBackground
-jelly: Jellyfish
-ps: ParticleSystem
-squid: PImage
-turtle: PImage
-seal: PImage
-seahorse: PImage

+setup(): void
+draw(): void
+mousePressed(): void
+keyPressed(): void
+keyReleased(): void
+playGame(): void
+drawStartScreen(): void
+handleStartScreenClick(): void

TankBackground

-waterTop: color
-waterBottom: color
-sandColor: color
-timeOffset: float
-generator: Random

+TankBackground(): void
+display(): void
+drawWater(): void
+drawMovingSand(): void

tentacle

-nodes: PVector [ ]
-numNodes: int
-stiffness: float

+Tentacle(x,y)
+reset(newX, newY)
+update(headX, headY)
+display()

Coral

-root: PVector
-angle: float
-aVelocity: float
-aAcceleration: float
-mass: float
-k: float
-damping: float
-h: float
-coralColor: color

+Coral(x: float, h: float): void
+update(externalForce: float): void
+display(): void

Jellyfish

-position: PVector
-pulseTime: float
-tentacleOffsets: ArrayList
-noiseSeed: float
-tentacles: Tent[ ]
-numTenctacles: int

+Jellyfish(x: float, y: float): void
+update(): void
+display(): void

ParticleSystem

-particles: ArrayList
-emissionRate: int
-originX: float
-originY: float
-position: PVector
-velocity: PVector
-lifetime: float

+update(): void
+ParticleSystem()
+addParticle(x,y)
+render(): void
+Particle(x,y)
+update()
+isDead(): boolean
+render()

Fish

-pos: PVector
-vel: PVector
-acc: PVector
-maxSpeed: float
-maxForce: float

+Fish(x: float, y: float): void
+applyBehaviours(fishes: ArrayList, cx: float, cy: float): void
+flee(target: PVector): PVector
+cohesion(fishes: ArrayList): PVector
+seek(target: PVector): PVector
+align(fishes: ArrayList): PVector
+separate(fishes: ArrayList): PVector
+update(): void
+display(): void